

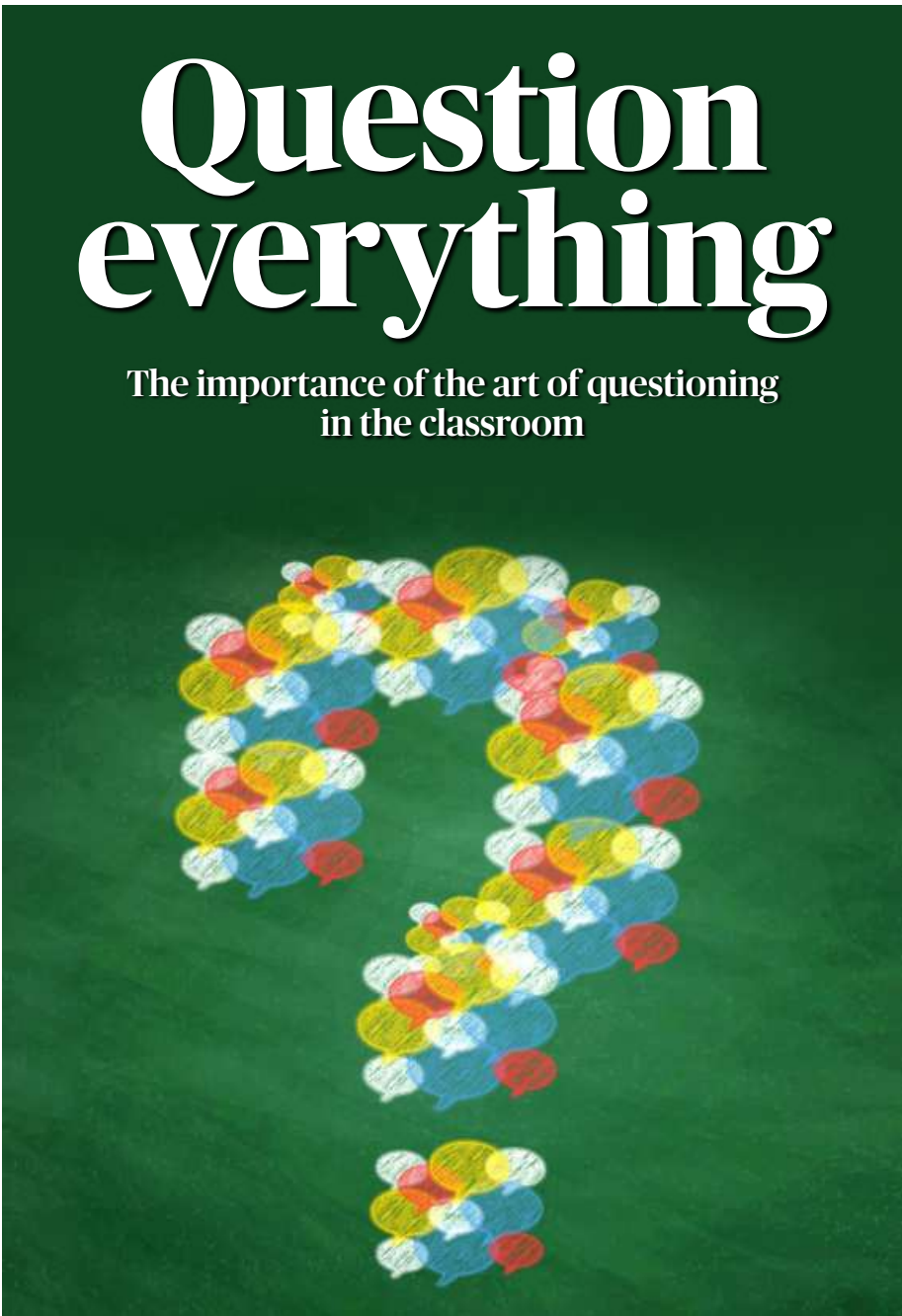
Sanil Raj Johnson

Some of the most significant discoveries in human history began with a simple question. The ability to ask thoughtful questions has always been central to intellectual growth, and classrooms that promote questioning often become spaces where genuine learning occurs. From the era of the Greek philosopher Socrates to today's research settings, questioning has remained a key method of inquiry.

Beyond testing

In contemporary classrooms, however, questioning often happens only after a lesson ends. Teachers typically ask a few predictable questions just to check if students have memorised the material. At a time when classrooms focus heavily on exams, documentation, and covering the syllabus, the art of questioning can revive curiosity and increase intellectual engagement.

Thoughtfully crafted questions activate a wide range of intellectual abilities and teach students to explore ideas from various perspectives and develop the skill to draw meaningful conclusions. Take, for example, a lesson on World War I. Instead of presenting the event as a collection of dates and facts, the teacher can guide the students through a sequence of questions. The first question might be: *Who were the two fac-*



tions involved? This can be followed by further inquiries: *How did the war begin? What were the political and economic causes be-*

hind it? What major events shaped its course? How did the conflict end, and what consequences did it have for the world?

As students attempt to answer these, they gradually construct the historical narrative themselves and understand the rela-

tionship among causes, events, and outcomes. Instead of memorising isolated facts, students begin to perceive history as a dynamic process shaped by interconnected factors.

The power of questioning extends across all academic disciplines. In Literature classrooms, for example, questions can guide students from basic comprehension to deeper interpretation. When introducing a poem, the teacher may begin with simple questions about the author and the title, and gradually, the discussion can move towards analytical inquiry. *What themes emerge from the text? What figures of speech are used? What message does the poet attempt to convey?*

Such questions encourage students to engage in discussion and enable them to participate actively in the process of interpretation.

In Science education, questioning plays a crucial role in sparking curiosity about the natural world. The history of Science shows how discoveries often start with simple questions. An apple falling from a tree led Isaac Newton to reportedly ask why the apple fell down. This curiosity eventually helped develop the law of gravitation.

In an elementary biology class, instead of merely describing the parts of a plant, a teacher might start by asking what role leaves play. As students move to higher levels, these questions can lead to more in-depth discussions about

plant biology and ecological systems.

Role in research

At advanced levels, questioning becomes the foundation of scholarly research. A doctoral thesis, for instance, is built upon carefully framed research questions. Scholars must define a thesis argument, formulate a hypothesis, justify the choice of topic, and explain the structure of their study. Questions regarding methodology, findings, and the scope for further investigation guide the research process. Thus, research itself can be understood as the systematic pursuit of answers to significant questions.

Yet, the most meaningful questions often come from the learner's own curiosity rather than those imposed from outside. They are not limited to textbook exercises or confined to the end of a lesson. Instead, they arise spontaneously whenever learners encounter something puzzling or intriguing. The teacher's role is not just to provide answers but to create an environment where questioning feels natural and ongoing.

Reclaiming the art and skill of questioning can be one of the most effective ways to promote deeper learning and sustained intellectual engagement.

The writer is a Professor of English and Dean of Student Affairs, Sahridaya College of Advanced Studies (Autonomous), Thrissur, Kerala. Email: saniljohn03@gmail.com

SCHOLARSHIPS

University of Bath International Excellence Scholarships

Offered by the University of Bath, the U.K.
Eligibility: New international applicants who hold an offer for a undergraduate course beginning in September and are assessed as overseas fee-paying students.
Rewards: £8,000 per academic year.
Application: Online
Deadline: June 25
www.b4s.in/edge/UBIEI

Dr. Rajendra Prasad Scholarship

An initiative of Buddy4Study.
Eligibility: Indian students across India who Class 10 or Class 12 in the 2025-26

academic year and scored minimum 65% in their previous class, and whose parents have an annual income not exceeding ₹800,000.

Rewards: Up to ₹30,000
Application: Online
Deadline: June 28
www.b4s.in/edge/DRPSP1

University of Hartford Future Leaders Scholarship

Offered by the University of Hartford, Connecticut, the U.S.
Eligibility: International students who have secured admission to a graduate programme and demonstrate required proficiency in English.
Rewards: \$2,000 a year
Application: Online
Deadline: July 15
www.b4s.in/edge/UHFL2

Courtesy: buddy4study.com

HCL Jigsaw

HCL's student innovation platform HCL Jigsaw invites registrations for its seventh edition. This year, the competition moves from traditional quiz-based models to teams that will work towards identifying real-world problems, building solutions, developing prototypes, and pitching their ideas at a national finale.

The themes are AI

and Digital Futures; Climate and Sustainability; Medical Innovation; Healthcare accessibility, smart diagnostics, wellness technologies, and assistive innovation; and Robotics and Automation.

Eligibility: Students from Classes 6 to 9 from any board or school across India. Registrations must be submitted through their respective schools only.

Deadline: July 31.
For details, visit www.hcljigsaw.com

Ask yourself questions

Uncertain about your career options? Low on self-confidence? This column may help



OFF THE EDGE
Nandini Raman

I am 23 and have a B.Sc. Physics and a Master's in Environmental Science. I am preparing for state government exams. I am worried about getting stuck. What if I end up doing nothing? Udhaya

Dear Udhaya,
Your Master's puts you in a fast-growing domain: Sustainability and Environmental, Social, and Governance (ESG). The private sector needs people who can bridge the gap between scientific data and corporate compliance. Consider roles such as conducting Environmental Impact Assessments (EIA) or calculating carbon footprint and resource efficiency to file a 'Business Responsibility and Sustainability Report (BRSR)' or ensuring that manufacturing plants meet pollution and safety standards.
Work on getting on some industry-standard certifications such as ISO 14001 Lead Auditor, GHG (Greenhouse Gas) Accounting, GIS and Remote Sensing. These will help make you employable if you decide against the exams. Also look at technical roles in government such as Scientific Assistant in the MoEFCC or CPCB. Niti Aayog hires Environmental Science postgraduates, as does ISRO.

I am an ECE graduate (2025 batch) working in a manufacturing company. My dream is to work in a marine industry. So I have quit and am preparing for

the role of Electro-Technical Officer in the merchant navy. How I can achieve my dream? Gangatharan

Dear Gangatharan,
First, ensure you meet the strict Directorate General of Shipping (DGS) standards for 2026. Do not join the ETO course without a job guarantee. Sponsorship is a legal agreement where a shipping company selects you before your training and guarantees you a four-month pre-sea course followed by eight months of onboard training. Top sponsorship companies are Anglo-Eastern (AEMA), Great Eastern (GEIMS), Mediterranean Shipping Company), Synergy Maritime/Fleet Management, and Shipping Corporation of India (SCI). The selection process involves an online written test, a psychometric test, and an interview.

After completing sea time, you have to appear for the Certificate of Competency (CoC) exam conducted by DG Shipping. Once you pass this, you are a certified Electro-Technical Officer. Apply for a Passport if you don't have one. You cannot get an INDOS number (seafarer ID) without it.

I have a Bachelor's degree in Zoology. I pursued law but took a break to prepare for the UPSC. I chose my degree to allow time for this. But I am worried about not having a backup. Naman

Dear Naman,
If you are close to finishing your Law degree, do it. A 'Science and Law' profile is recession-proof. Dedicate 80% of your day to UPSC and 20% to a skill like

Environmental Audit or IPR Law. Look at a short-term certification in Environmental Impact Assessment (EIA), as it makes you employable in any major infrastructure firm.

Some other career paths are Environmental Law and ESG. Since you can understand animal ecosystems (Zoology) and legal frameworks of the National Green Tribunal (Law), consider Wildlife Forensics and Crime Investigation. The Wildlife Institute of India (WII), Dehradun, and the Zoological Survey of India (ZSI) hire for project-based roles that can transition into permanent government positions. Law firms and pharmaceutical companies are always looking for 'Patent Associates' who have a Science background to become a Patent Agent or Intellectual Property (IPR) Consultant. You will need to take the Indian Patent Agent Examination.

Since you are already in 'study mode,' pivot to exams where your Zoology degree gives you an edge for the non-UPSC technical Government jobs.

I'm pursuing a B.A. (Hons.) with Research in Political Science and International Relations. I would like to do a Master's in Human Rights and Humanitarian Action from Sciences Po University. What are my options? How should I approach the scholarship application? Niranjana

Dear Niranjana,
You seem to be looking at a career in direct field intervention, policy advocacy, or international protection. Graduates from this Master's course typically split into Humanitarian

Action and Human Rights programmes.

Humanitarian action roles such as Emergency Response Coordinator, Food Security Officer, Refugee Protection Officer are key in organisations such as the International Committee of the Red Cross, Doctors Without Borders, the UN World Food Programme (WFP), and UNHCR.

Human rights roles involve being a Human Rights Investigator, Policy Advocate, Legal Consultant, and Research Analyst across organisations such as Amnesty International, Human Rights Watch, Office of the UN High Commissioner for Human Rights, and regional bodies such as National Human Rights Commission in India.

Internships with organisations working with refugee integration, manual scavenging, or tribal rights will strengthen your profile. Consider a stint at the NHRC or NITI Aayog (Social Justice vertical), or think tanks such as ORF and IDSA. Aim for publications in peer-reviewed student journals like the *Journal of International Relations* or the *Cornell International Affairs Review* and on platforms such as *The Quint*, *The Wire*, or *Modern Diplomacy*. Ensure that your focus area of research is not generic. For an Indian student, there are three main funding avenues for Sciences Po: Émile Boutmy, Charkap (Master's) and the Inlaks Foundation scholarship. Apply for the scholarship when you apply for admission. Your SOP must connect your past research to your future goal of helping people.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge



Lay a strong foundation

Why fieldwork is the finest teacher for students of Civil Engineering and Construction Management

Phani Madhavi Talasila

Leadership in Civil Engineering and Construction Management is not born in air-conditioned classrooms. It is shaped under the open sky, amid dust, deadlines, drawings, and decisions. For students aspiring to become civil engineers and construction managers, fieldwork is not merely a curriculum requirement but is the foundation of true leadership and infrastructure excellence.

In classrooms, students learn and understand theories of structural design, construction planning and scheduling, project management, construction materials, codal provisions and specifications. However, the construction site imparts something far more powerful. When a student steps onto a construction project site, the drawings have evolved into real foundations, columns, beams, and slabs. This is where knowledge meets reality. How does fieldwork contribute to developing leadership?

Gateway to real-world challenges: No textbook can fully replicate the unpredictable conditions of weather changes, labour management issues, supply delays, equipment breakdowns, or coordination conflicts between various teams. On site, decisions must be practical, quick, and effective. At this phase, students begin to understand that leadership is not about authority, but about accountability.

Communication is the core: A civil engineer on site must coordinate with architects, structural consultants, contractors, safety officers, and workers. Students quickly realise that technical knowledge alone is not enough. They must listen carefully, explain clearly, and resolve conflicts diplomatically. These experiences shape confidence and maturity.

Responsibility and discipline: Construction projects operate on strict timelines and budgets. Even minor delays and errors lead to financial constraints and can threaten safety. When students wit-

ness how a small mistake in measurement can affect an entire structure, they learn the importance of precision and attention to detail. Leadership grows from this sense of responsibility.

Safety management: Hard hats, safety harnesses, barricades, and safety briefings are life-saving measures. Students who participate in safety management develop a profound respect for human life and professional ethics and learn that a true leader prioritises safety above speed or cost.

Problem-solving skills: Unexpected soil conditions, design changes, material shortages, or coordination gaps demand immediate solutions. Students learn to think critically, evaluate alternatives, and take calculated decisions. They understand that leadership often means making tough calls under pressure.

Understanding teamwork: From skilled labourers to site engineers and project managers, every individual contributes to the

final outcome. Students who work closely with site teams learn how practical experience complements theoretical expertise and realise that leadership is about guiding teams, not dominating them.

Live laboratory: For construction management students, the site becomes a live laboratory of planning and execution. Gantt charts and CPM networks studied in class come alive in real progress meetings. Cost estimation exercises translate into real procurement negotiations. Quality control procedures become site inspections. The gap between theory and practice narrows, and confidence grows.

Builds resilience: Construction sites are demanding environments; long hours, dynamic challenges, and constant monitoring test patience and endurance. Students who endure these conditions develop strength of character, stamina and determination.

Classroom of ethics: Transparency in measurements, fairness in labour practices, adherence to quality standards, and honesty in reporting ... these values are observed, practised, and internalised on site. Field exposure ensures that students understand professional responsibility beyond textbooks.

While automation and smart tools have enhanced site operations today, field presence remains irreplaceable. Students must experience actual site work to become leaders capable of managing people and technology effectively. Universities may provide knowledge but fieldwork transforms students into professionals who can plan intelligently, execute efficiently, communicate effectively, and lead responsibly.

The writer has a Ph.D. in Civil Engineering and writes on sustainable infrastructure, environmental sustainability, and higher education.

Aritra Ghosal

The recent India-New Zealand Free Trade Agreement should not be read only as a trade development. Its significance for international education lies in the formal inclusion of student mobility within a bilateral framework. At a time when visa systems across major destinations are becoming more selective and less predictable, this agreement signals a different approach. Student movement is no longer being treated as a stand-alone policy lever. It is being positioned within a broader, long-term alignment between education, workforce needs, and skilled migration.

Structured understanding

One of the most important elements is the introduction of a dedicated annex on student mobility and post-study work, something New Zealand has not

previously formalised with any country in this manner. This changes the nature of the student movement. Instead of being governed entirely by immigration rules that can evolve with domestic priorities, it is now supported by a structured bilateral understanding. The difference is not procedural. It is directional. For students, this reduces the policy uncertainty that has become increasingly common over the last few years.

The agreement provides a clear articulation of work rights both during and after study. Students are permitted to work up to 20 hours week a during their academic programme, with the added assurance that this provision will continue even if policy settings change in the future. This reflects an attempt to build predictability into the system. Post-study work timelines are also explicitly defined. Students completing STEM undergraduate programmes can stay back for



GETTY IMAGES/ISTOCKPHOTO

up to three years. Those pursuing Master's degrees are eligible for similar durations while doctoral graduates can access extended work opportunities

of up to four years. The reason these provisions are different lies in the context in which they are embedded.

The agreement does not

stop at post-study work. It introduces a structured transition into employment through a Temporary Employment Entry pathway, with a defined

quota of 5,000 visas for skilled professionals. This is where the framework moves beyond education. The sectors covered under this pathway – technology,

engineering, healthcare, education, and construction – are aligned with areas of demand. This creates a more direct relationship between what students study and where opportunities exist

For Indian students, frameworks such as this push decision-making towards greater alignment. Course selection, in particular, becomes more closely linked to employability and sector demand. The value of a programme is now tied to how effectively it connects to workforce participation. This is a shift from aspiration-led choices to outcome-linked decisions.

Limits

It is equally important to recognise the limits of what the agreement offers. Structured pathways do not eliminate competition. Employment outcomes will continue to depend on labour market conditions, employer absorption, and individual preparedness. Migration routes, even when formalised, operate

within quotas and sectoral priorities. The presence of a pathway does not ensure progression through it. There is also the question of execution. Policy intent provides direction but on-ground outcomes are shaped by implementation, processing timelines, institutional quality, and economic demand.

The India-New Zealand FTA does not make studying abroad more defined. For students, this introduces both clarity and expectation. Academic choices will need to align more closely with skill demand. Applications will need to demonstrate clearer intent. Long-term outcomes will depend less on destination preference and more on how effectively students navigate structured pathways. That is the direction international education is moving towards. The advantage will lie with those who recognise it early.

The writer is Founder and Director of OneStep Global.

Beyond promises

Rethinking the role of Education ministers in India



WIDE ANGLE
Albert P' Rayan

Recently, while preparing a talk on the education-related promises made by political parties in Tamil Nadu, I read the manifestos of several established parties as well as a new entrant. I was genuinely impressed by the importance they attached to education.

Ambitious vision

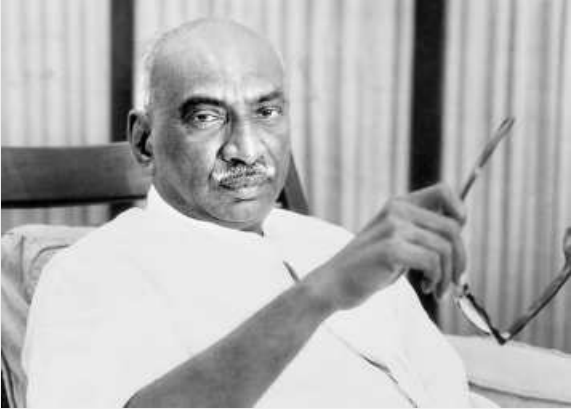
Most parties present ambitious visions for the sector. Their proposals include technology-driven learning, skill development, education loans, and job-oriented training. At the same time, they emphasise inclusive access, welfare measures, financial assistance, dropout prevention, and affordable higher education, especially for socially and economically disadvantaged groups. Taken together, these ideas reflect a common recognition that education is not only a basic right but also a powerful means of social and economic progress.

As Keats wrote in *Ode on a Grecian Urn*, "Heard melodies are sweet, but those unheard are sweeter." Political promises, too, can sound attractive and inspiring. Yet what follows after a party comes to power is not always sweeter; at times, it can be quite the opposite.

If implemented sincerely, however, these promises could make Tamil Nadu a model for the rest of India. This raises a crucial question: who will head the Ministry of Education? Political parties rarely tell voters who will serve as Education Minister if they came to power.

Recent developments have made this question even more significant. The appointment of Mr. A. Rajmohan as Tamil Nadu's School Education Minister sparked controversy over his educational qualifications.

At the national level, several opposition leaders and members of the public have demanded the resignation of Union Education Minister Mr. Dharmendra Pradhan due to the cancellation of the NEET-UG 2026 following alleged leaks of the



question paper and criticism of the government's handling of exam-related irregularities.

Qualities required

We often discuss the qualifications and qualities required to become a teacher and the role teachers play in shaping society. However, rarely do we reflect on the qualities required of an Education Minister or the role such leaders play in shaping society through policymaking. Nor do we sufficiently examine what contemporary education leaders can learn from outstanding Education Ministers in India and abroad.

What makes a good Education Minister: broad vision or high academic qualifications? This is a false binary. In today's world, the person heading the Ministry of Education must not only possess sound academic understanding but also administrative competence, integrity, and a forward-looking vision. Those who downplay the importance of formal qualifications often cite the example of former Tamil Nadu Chief Minister K Kamaraj, who reportedly studied only up to Std. VI. Yet he is remembered as one of Tamil Nadu's greatest leaders because he understood the transformative power of education and worked tirelessly to expand access to schooling. His life demonstrates that vision, commitment, and leadership can sometimes matter more than degrees alone.

Complex challenges

At the same time, the challenges facing education today are far more complex. Policymakers must grapple with issues such as educational equity, curriculum reform, digital learning, and the changing demands of society.

To lead effectively in such a context, an Educa-

tion Minister must combine intellectual competence with integrity, administrative skill, and a deep commitment to public education.

A good Education Minister should understand society's long-term needs, implement reforms effectively, ensure that education reaches rural and marginalised children, respect educators and experts, consult them meaningfully, and translate ideas into functioning systems.

Two names immediately come to mind as inspiring examples of education leaders who used education as a tool for social transformation. Maulana Abul Kalam Azad, India's first Education Minister, laid the foundations of the nation's academic and scientific infrastructure and played a key role in establishing institutions such as the IITs and the UGC. Paulo Freire, best known for Pedagogy of the Oppressed, served as Education Secretary in São Paulo, Brazil, where he attempted to translate his philosophy of critical pedagogy into concrete educational reforms. Both demonstrated conviction, commitment and courage.

An Education Minister is not just a political figure, but someone responsible for shaping the future of a generation.

At a time when society is facing rapid technological changes, growing inequality, and complex educational challenges, India needs Education Ministers who possess vision, empathy, academic knowledge, administrative skill, scientific temper and a genuine interest in education. Only such leadership can make education a true force for social progress.

The writer is an ELT resource person and education columnist. Email rayanal@yahoo.co.uk

The next in the monthly series by WWF-India that highlights niche and unconventional green careers through the stories of well-known personalities from the field of environment and conservation

Most people have one career but I've had about five. Journalist, investment banker, software company manager, children's newspaper editor, and now natural history publisher. The last one arrived unexpectedly, but it feels most like home.

Today, I run *Indian Pitta*, a nature imprint published by Juggernaut, where I edit and publish books on wildlife, birds, and conservation. But the journey here was anything but straight. I think that is precisely the point. You do not have to follow a conventional path into a green career.

I grew up in Chennai listening to stories about my grandfather, a journalist who worked with the *Free Press Journal* in Bombay during the years when India won its independence. After a postgraduate programme in Mass Communication in Bombay, I joined *The Hindu Business Line*, moved into invest-

GREEN CAREERS HUB Words, wings, and a winding road

Writer, editor, and publisher
Anita Mani on her journey from newsrooms to Nature

ment banking, and later into software operations. The industries changed, but communication remained at the centre of my work. Those years also exposed me to extraordinary landscapes and wildlife.

Finding my niche

The passion for Nature and birdwatching that now defines my work entered my life through an unexpected place: my son. He became fascinated with birds at a young age. As he was too young to travel alone, I drove him to wetlands, lakes, and national parks around Delhi almost every weekend. Somewhere along the way, I stopped waiting in the car and started paying attention. That was nearly 20 years ago!

At the same time, I ran



Child Friendly News, a children's newspaper. When the pandemic shut it down, I realised I had spent years building the

YOUNG ACHIEVER

Design meets diplomacy

A student from India on representing the country at the 2026 ECOSOC Youth Forum in the U.S.



mate Ownership', highlighting that youth in the Global South do not need reminders of the impending crisis. Instead, they need the technical tools to solve it.

To illustrate this, I shared a project in which I re-engineered a traditional manual process, reducing operational time from 12 hours to just 15 minutes. It demonstrated that applying industrial design to systemic problems can help make the UN's Sustainable Development Goals (SDGs) more achievable.

Lessons learnt

One of my biggest takeaways was seeing how much the world is watching India's digital growth, especially while interacting with the delegates from Malawi. From high-level technology to governance, healthcare, digital infrastructure and their people, we spoke about everything. The conversation highlighted how a system is only as effective as it is accessible.

In design, we call this 'User Experience'. At the UN, I realised that public policy is just as similar. If the 'user' (the citizen) finds it too complicated to use, the policy fails. I realised that designing a clear, accessible government system is just as vital as designing a physical tool.

Another important lesson was the contrast bet-

ween the pace of diplomacy and that of design. While diplomacy works deliberately and decisively with every word measured and every decision matured at its own time, design thrives on iterations: build, fail, fix, and improve.

Bridging these two worlds was a challenge, but also an opportunity. My role as an envoy was to act as a translator, turning complex ideas from design labs and rural fieldwork into a narrative that moves diplomacy. I learnt that, to change the world, you must explain the 'how' with the precision of a designer and the 'why' with the conviction of a diplomat.

This experience has fundamentally reshaped my perspective. I no longer see industrial design merely as the creation of objects, but as the creation of systems that uphold human dignity. I now wish to focus on Technical Diplomacy, ensuring that the new world we build is designed with humans at the centre.

Representing India also underscored a larger shift. We are no longer just participating in global conversations; we are shaping them. The conference may be over, but my work continues.

The writer is a student of B.Des. Industrial Design at Anant National University.